

Tagged Prime-Ideal Packets and the Rational Pushforward Obstruction for Hénon Monodromy

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Abstract

Full multiplier-field norms of primitive cyclotomic Hénon packets are squares, but their inversion-fixed trace-field factors and prime ideals remain source-native arithmetic objects. We construct an exact finite-cutoff assembly of those survivors. Each atom retains the signed primitive orbit, cyclotomic index, trace-field prime ideal, residue degree, and ideal valuation. The rational norm is an exact pushforward of this tagged divisor. Moreover, at residue characteristic not dividing the cyclotomic index, every supporting multiplier-field prime sees the unstable multiplier with exact residue order equal to that index. An exact certificate for signed primitive H6 periods one, three, and four and indices $3 \leq n \leq 20$ contains 125 tagged atoms over 95 rational primes. The free rational pushforward therefore has kernel rank 30. Explicit collisions show that the same rational prime carries different orbit labels, different indices, different trace primes, and even incompatible exact residue orders; $p = 29$, for example, carries orders 7, 14, and 15. Thus a rational-prime-only ledger is not a lossless H6 Euler clock. The obstruction is scoped: weighted scalar statistics may remain useful, but an all-orbit trace first requires a convergent vector-valued tagged assembly. No analytic continuation or Hilbert–Pólya operator is claimed.

1 Introduction

Periodic-orbit constructions often turn local determinant data into scalar Euler factors. For the certified area-preserving H6 Hénon survivor, this strategy encounters a sequence of algebraic obstructions. The primitive return matrices have reciprocal algebraic-unit multipliers, but no common real power maps the exact primitive controls to rational primes. Passing to cyclic resultants does produce integral sequences; taking the full multiplier-field norm, however, counts the inversion pair twice and yields a square. The smallest surviving arithmetic objects are therefore the unit-normalized cyclotomic values in the fixed trace field and their prime-ideal factorizations.

Cyclic resultants and their recurrences are well developed for a fixed polynomial [3, 4]. Primitive divisor theorems also give strong information for Lucas, Lehmer, and certain quadratic Lehmer–Pierce sequences [1, 2]. Arithmetic dynamics supplies related Zsigmondy phenomena for iterates of a fixed rational map [5, 6]. None of these results provides a common divisor ledger for varying signed H6 monodromy fields and pressure-weighted primitive orbits.

This paper takes one deliberately finite but exact step. We define a free tagged divisor group whose basis remembers an orbit, a cyclotomic index, and a trace-field prime ideal. We prove that rational norms are exact residue-degree pushforwards of this object. We then attach an intrinsic clock: away from residue characteristics dividing the index, every multiplier-field prime above a packet atom gives exact multiplier order. Finally, an executable H6 certificate proves that forgetting to rational primes is noninjective.

The result has two complementary messages. Positively, the phrase “retain the prime ideals” becomes a concrete finite data type with exact functoriality. Negatively, a bare rational prime cannot be a lossless local label: it can support incompatible source clocks. This does not rule out derived scalar statistics. It says that scalar pushforward must be treated as a lossy map, not as an identification.

2 Signed H6 fields and half packets

Consider

$$H_6(q, p) = (1 - 6q^2 - p, q), \quad J(q) = \begin{pmatrix} -12q & -1 \\ 1 & 0 \end{pmatrix}.$$

Every chronological return matrix has determinant one. For a primitive orbit γ , let λ_γ denote the actual signed unstable eigenvalue. Its reciprocal is the second eigenvalue. The sign is part of the source data: the exact period-three branch is negative.

The three controls used below have multiplier polynomials

$$f_1(X) = X^4 - 4X^3 - 22X^2 - 4X + 1, \quad (1)$$

$$f_3^-(X) = X^4 + 76X^3 - 7374X^2 + 76X + 1, \quad (2)$$

$$f_4(X) = X^2 - 578X + 1. \quad (3)$$

They are reciprocal and monic with constant term one. Write $K_\gamma = \mathbb{Q}(\lambda_\gamma)$, let ι be inversion, and set $F_\gamma = K_\gamma^{\langle \iota \rangle}$. Their fixed trace fields and trace values $t_\gamma = \lambda_\gamma + \lambda_\gamma^{-1}$ are

primitive period	F_γ	t_γ
1	$\mathbb{Q}(\sqrt{7})$	$2 + 2\sqrt{7}$
3	$\mathbb{Q}(\sqrt{5})$	$-38 - 42\sqrt{5}$
4	$\mathbb{Q}$	578.

For $n > 2$, define the inversion-fixed half packet

$$\beta_{\gamma,n} = \lambda_\gamma^{-\varphi(n)/2} \Phi_n(\lambda_\gamma). \quad (4)$$

Since $\varphi(n)$ is even, cyclotomic reciprocity shows that $\iota(\beta_{\gamma,n}) = \beta_{\gamma,n}$. The multiplier is a unit, so $\beta_{\gamma,n} \in \mathcal{O}_{F_\gamma}$. This is the trace-field survivor of the full-field square norm.

3 Finite tagged assembly

Let S be a finite set of pairs (γ, n) with $n > 2$. For every supporting trace-field prime define a formal source atom.

Definition 3.1 (Tagged packet divisor). Let

$$\mathcal{A}_S = \{(\gamma, n, \mathfrak{q}) : (\gamma, n) \in S, \mathfrak{q} \mid (\beta_{\gamma,n})\}$$

and

$$\text{Div}_{\text{tag}}(S) = \bigoplus_{(\gamma,n,\mathfrak{q}) \in \mathcal{A}_S} \mathbb{Z}[\gamma, n, \mathfrak{q}].$$

The finite tagged divisor is

$$\mathcal{D}_S = \sum_{(\gamma,n) \in S} \sum_{\mathfrak{q}} v_{\mathfrak{q}}(\beta_{\gamma,n}) [\gamma, n, \mathfrak{q}]. \quad (5)$$

Here canonical means canonical relative to the fixed signed monodromy and the finite source set S . We do not claim universal minimality among all possible encodings.

Let $p = \mathfrak{q} \cap \mathbb{Z}$ and let $f(\mathfrak{q}/p)$ be the residue degree. Define

$$\nu_S : [\gamma, n, \mathfrak{q}] \mapsto f(\mathfrak{q}/p)[p] \quad (6)$$

from the tagged group to the free rational-prime divisor group.

Theorem 3.2 (Exact finite assembly and norm pushforward). *The divisor (5) is a lossless finite record of the source prime-ideal factorizations. For every $(\gamma, n) \in S$,*

$$\operatorname{div}_{\mathbb{Z}} \left| N_{F_{\gamma}/\mathbb{Q}}(\beta_{\gamma,n}) \right| = \sum_{\mathfrak{q} | (\beta_{\gamma,n})} v_{\mathfrak{q}}(\beta_{\gamma,n}) f(\mathfrak{q}/p)[p]. \quad (7)$$

Consequently the rational divisor of the product of all packet norms is $\nu_S(\mathcal{D}_S)$.

Proof. Unique factorization of nonzero ideals gives $(\beta_{\gamma,n}) = \prod_{\mathfrak{q}} \mathfrak{q}^{v_{\mathfrak{q}}}$. The ideal norm of $\mathfrak{q}$ is $p^{f(\mathfrak{q}/p)}$. Taking norms and then rational prime divisors proves (7). The source tags are basis coordinates in a finite free group, so they are retained before the pushforward. Linearity proves the final assertion. $\square$

Remark 3.3 (Where the content lies). The retention of the tags is definitional once $\operatorname{Div}_{\operatorname{tag}}(S)$ is chosen. The substantive assertions are that the H6 half packets canonically supply its coefficients, that residue degrees give the exact norm map (7), and that the same atoms carry the good-characteristic order in Theorem 4.1. No novelty or minimality claim is made for free divisor groups in general.

4 The intrinsic good-characteristic clock

A trace-field atom supports primes in the multiplier field. These two prime levels must be kept distinct.

Theorem 4.1 (Exact residue order). *Let $\mathfrak{q} | (\beta_{\gamma,n})$ and let $\mathfrak{P}$ be any prime of K_{γ} above $\mathfrak{q}$, of residue characteristic p . If $p \nmid n$, then the reduction of λ_{γ} in $(\mathcal{O}_{K_{\gamma}}/\mathfrak{P})^{\times}$ has exact multiplicative order n .*

Proof. The unit factor in (4) implies $(\beta_{\gamma,n})\mathcal{O}_{K_{\gamma}} = (\Phi_n(\lambda_{\gamma}))$. Hence $\Phi_n(\lambda_{\gamma}) \in \mathfrak{P}$. Write $\bar{\lambda}$ for the reduction. Since $p \nmid n$, $X^n - 1$ is separable in characteristic p . In the reduced factorization

$$X^n - 1 = \prod_{d|n} \Phi_d(X),$$

the monic factors are therefore pairwise coprime. Now $\Phi_n(\bar{\lambda}) = 0$ implies $\bar{\lambda}^n = 1$. If its order were a proper divisor $d < n$, then $\bar{\lambda}$ would also be a root of $X^d - 1 = \prod_{e|d} \Phi_e(X)$, contradicting that coprimality. Its order is exactly n . $\square$

Remark 4.2 (Bad characteristic). When $p | n$, the reduced cyclotomic factors need not remain coprime and the order can drop. Such atoms are retained in the divisor ledger but are not certified by Theorem 4.1.

Thus n is not merely a bookkeeping index at good characteristic. It is an intrinsic finite-field clock. Any pushforward that merges atoms with different n can merge different exact clocks.

5 The rational pushforward obstruction

The norm map in Theorem 3.2 is exact but need not be injective. The following rank formula makes the information loss explicit.

Proposition 5.1 (Free kernel rank). *Let $m = \#\mathcal{A}_S$, and let r be the number of distinct rational primes below atoms in $\mathcal{A}_S$. Then*

$$\text{rank}_{\mathbb{Z}} \ker \nu_S = m - r.$$

In particular, $m > r$ implies noninjectivity.

Proof. For a fixed rational prime p , suppose m_p tagged atoms lie above it. The p -block of (6) is

$$\mathbb{Z}^{m_p} \longrightarrow \mathbb{Z}, \quad (x_i) \longmapsto \sum_i f_i x_i,$$

with all $f_i > 0$. Its rank is one and its free kernel has rank $m_p - 1$. Summing these independent blocks over the r rational primes gives $\sum_p (m_p - 1) = m - r$. $\square$

Noninjectivity is a scoped obstruction. It does not prohibit a useful weighted pushforward; it prohibits calling the untagged rational divisor a lossless reconstruction of the source packet.

6 Exact H6 certificate

The certificate computes the half packets for the signed controls (1)–(3) and indices $3 \leq n \leq 20$. It reconstructs the trace polynomials, uses exact integral bases, factors ramified, inert, and split trace primes, and checks the multiplier order in every supporting finite-field factor. No floating-point approximation or external prime table is used.

The finite ledger contains

54 packet rows, 125 tagged atoms, 95 distinct rational primes.

Proposition 5.1 therefore gives

$$\text{rank} \ker \nu_S = 125 - 95 = 30.$$

This is not merely a signed-group artifact. At $p = 109, n = 11$, three distinct effective basis atoms all have residue degree one and hence each maps to the same effective divisor [109]. Noninjectivity is already visible on the positive atom set. There are 105 good-characteristic atoms with exact order and 20 bad-characteristic atoms deliberately left uncertified. All 54 rational norm factorizations reconstruct exactly, including 30 inherited half-norm crosschecks. The canonical core digest is

9cc40f4b8ed5f54e36c70cb1db15b509837aee264e86bcf39395f827e59153ba.

Table 1 shows decisive source collisions.

The table exposes three independent tag requirements on this certificate. At $p = 109, n = 11$, periods one and three coexist, so the orbit tag is needed. Within period one, two distinct split trace primes coexist, so the prime-ideal tag is needed. At $p = 29$, the certified orders 7, 14, and 15 coexist, so retaining only the rational prime loses the cyclotomic index and clock. Further multi-order collisions occur at

$$p \in \{11, 19, 29, 79, 131, 307, 38039\}.$$

The signed period-three branch is separately mutation-tested: replacing it by the positive modulus changes its packet.

These controls do not prove that our key is categorically minimal. They prove that the displayed orbit, index, prime-ideal, and sign coarsenings lose actual information in the exact H6 data.

Table 1: Selected tagged atoms. The order column is the exact multiplier residue order; all displayed rows are good characteristic.

p	orbit	n	trace prime ideal	f	v_q	order
29	γ_1	15	(29,w-6)	1	1	15
29	γ_3^-	7	(29,w-24)	1	1	7
29	γ_3^-	14	(29,w-6)	1	1	14
109	γ_1	11	(109,w-15)	1	1	11
109	γ_1	11	(109,w-94)	1	1	11
109	γ_3^-	11	(109,w-99)	1	1	11
131	γ_1	12	(131,w-111)	1	1	12
131	γ_3^-	5	(131,w-120)	1	1	5
38039	γ_3^-	19	(38039,w-22136)	1	1	19
38039	γ_4	13	(38039) in $\mathbb{Z}$	1	1	13

7 Source boundary and the next theorem

Hillar’s cyclic-resultant framework and the Hillar–Levine recurrence theory concern one fixed polynomial [3, 4]. Flatters’ theorem supplies eventual primitive rational divisors for the period-four real-quadratic Lehmer–Pierce sequence [2], while the Bilu–Hanrot–Voutier theorem treats Lucas and Lehmer numbers [1]. Ingram–Silverman and related dynamical-Zsigmondy work treat orbits of a fixed rational map [5, 6]. These are strong positive analogues, but none accepts the present family of varying signed H6 monodromy fields and pressure weights.

The next theorem must therefore begin with the tagged source object. A schematic target is a pressure-filtered divisor measure

$$\mathfrak{D}_T = \sum_{\substack{\gamma \text{ primitive} \\ \ell_P(\gamma) \leq T}} W_T(\gamma) \sum_{n \geq 3} \sum_{q | (\beta_{\gamma,n})} v_q(\beta_{\gamma,n})[\gamma, n, q].$$

One must specify a common vector-valued topology, derive W_T from the H6 pressure clock, and prove convergence before applying a rational norm map. No such theorem is supplied here.

The strict Route-A tuple is

$$(A1_{\text{WEAK}}, A2_{\text{FAIL}}, A3_{\text{FAIL}}, A4_{\text{FORMAL_HINT}}), \quad \text{ROUTE_A_EXPLORATORY}.$$

There is no global Fredholm determinant, continuation theorem, functional equation, Hilbert space, or self-adjoint operator. Route B is not authorized.

8 Conclusion

Prime-ideal packets can be assembled across a finite signed H6 source set without losing their provenance. Rational norms are exact pushforwards, and good-characteristic packet atoms carry the cyclotomic index as an exact residue-order clock. The same exactness reveals the obstruction: the rational-prime-only map has a positive free kernel and merges incompatible

clocks. The mathematically honest next bridge is therefore vector-valued and pressure weighted. Scalarization, if useful, must come after that assembly theorem rather than replace it.

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